Metabolic Parameters of the Earthworm *Eisenia foetida* in Relation to Temperature

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Summary

Comparisons of rate of achieving sexual maturity, cocoon production, and growth rate at six temperatures between 15 and 30°C led to the conclusion that the earthworm *Eisenia foetida* can be produced optimally over the range 18-25°C and perhaps maximally at 20°C.

INTRODUCTION

Despite suggestions that *Eisenia foetida* (Savigny) may be useful for management of certain organic wastes[1-4] and for production of protein,[5-7] optimum temperature for total biomass production by this earthworm over a long period is not known. Working with isolated individuals, Tsukamoto and Watanabe[8] showed that growth from hatchling to adult occurred more rapidly at 25°C than at 20°C. Data of Kirberger[9] indicate a zone of metabolic thermal compensation between 15 and 20°C for fed, but not fasted *E. foetida*, provided the worms had been acclimated at 15°C and not 25°C. In contrast, data of Mitchell[10] show that this zone lies between 20 and 25°C. Mitchell's finding was confirmed by Moment and Habermann[11] who also showed that this finding is independent of acclimation temperature. Reinecke and Kriel[12] obtained 194 cocoons in 20 days from *E. foetida* at 25°C compared to 106 cocoons at 20°C, but only 2.5 hatchlings per cocoon at 25°C compared to 3.1 at 20°C; since 98% of 100 cocoons hatched at 20°C and 94% at 25°C, a total of 322 progeny were derived at 20°C versus 456 at 25°C.

The objectives of this study were to determine survival, onset of hatching, onset of sexual maturity, fecundity, growth rate, and total biomass production in relation to both population density and temperature, and to determine the optimum temperature at which these phenomena, collectively, would work to maximize biomass production over a period in excess of the interharvest period generally used by commercial growers.[7]

METHODS AND MATERIALS

Activated sludge was obtained and dewatered, and earthworm hatchlings collected, as described earlier.[13] A single hatchling, or populations of 2, 4, or

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8, were placed into 20 × 100 mm Petri dishes with 30 g of sludge [12–15% solids (w/w)]. Five replicates of each population density were placed in constant temperature (±1°C) cabinets at 15, 18, 20, 25, 28, and 30°C. The earthworms were weighed and transferred to fresh medium at 30, 50, 65, and 80 days. Prior to transfer, each was examined to determine whether a clitellum was present, denoting sexual maturity. The spent medium was searched for cocoons, which were counted and removed from the experiment. An analysis of variance was applied to the data, and significant values were distinguished by the procedure of Keul.[14]

RESULTS

Survival

All of the earthworms survived 80 days at 15°C (Table I). At 18, 20, and 25°C, mortality occurred only in dishes with eight earthworms, but at 28°C, among populations of four as well. At 30°C, mortality occurred even in dishes which contained a single worm.

Onset of Sexual Maturity

None of the 75 hatchlings at 15°C became capable of reproducing until some time after seven weeks, though by this time about one-third of the individuals at 18°C and two-thirds at 20, 25, and 28°C displayed a clitellum (Table II). Before an age of 65 days all, or nearly all, of the worms were sex-

TABLE I
Percent Survival of *E. foetida* in Relation to Time, Temperature, and Population Density[a]

Temperature (°C)	Population density	Time (days)			
		30	50	65	80
18	8	100	80	80	40
20	8	100	100	100	82
25	8	95	92	92	90
28					
	4	100	90	85	80
	8	100	90	72	42
30					
	1	100	80	80	60
	2	100	50	50	30
	4	100	100	90	20
	8	80	37	37	15

[a]Values based on initial numbers of 5, 10, 20, and 40 individuals in dishes with 1, 2, 4, or 8 earthworms, respectively. Survival was 100% at 15°C for all population densities, and 100% at other temperatures for population densities not listed.

TABLE II
Percent Clitellate *E. foetida* in Relation to Temperature and Time

Temperature (°C)	Time (days) 30	50	65	80
15	0	0	15	99
18	0	0	99	100
20	1	63	100	100
25	28	71	100	100
28	28	62	95	97
30	6	8	12	36

ually mature at 18–28°C, whereas only one-third of the 16 survivors at 30°C were mature after 80 days.

Cocoon Production and Development

None of the earthworms at 30°C and few at 15°C were fecund after 80 days, while a similar and maximum number of cocoons was produced at 20 and 25°C (Table III). The rate of development of progeny, terminating with hatching from cocoons, was proportional to temperature (Table IV).

Growth

Results on day 30 suggested that growth would occur more rapidly at 25 or 28°C than at 20°C (Table V). Results obtained with singly grown worms or populations of two or four worms after 50 days, however, suggested that growth was statistically equal at 20, 25, and 28°C; populations of eight worms exhibited statistically lower mean weights, indicating an inadequate food supply (Table VI). After 80 days all worms in populations that had survived 15,18, 20, 25, and 28°C were statistically equal in weight, while singly

TABLE III
Cocoons Produced by *E. foetida* in Relation to Temperature and Time

Temperature (°C)	Time (days) 30	50	65	80	Totals
15	0	0	0	11	11
18	0	0	56	20	76
20	0	15	230	9	254
25	0	32	180	6	218
28	0	10	60	2	72
30	0	0	0	0	0

TABLE IV
Percent Hatching of *E. foetida* in Relation to Temperature

Temperature (°C)	Cocoons tested[a] and time interval (days between setup and analysis)[b]					
	113(21)	80(19)	22(19)	54(16)	30(7)	$\bar{x}$
20	45	20	45	24	0	27
25	54	35	45	31	13	36
30	67	42	44	37	20	42

[a]Cocoons were obtained at intervals of 7-10 days of setting up cultures and were randomly distributed among dishes at the three test temperatures.
[b]Attempts were made to determine percentage when percent differences may have been greatest between temperatures.

grown worms at 15°C and all worms at 30°C were significantly lighter (Table VII).

Data given in Table VIII show an apparent progression in growth rate from a maximum at 25-28°C after 30 days to a broader maximum at 20-28°C after 30 days, to a shift of this maximum to the interval 15-25°C after 65 and 80 days. Slopes for regression analysis of these data show an apparently similar rate of growth of the worm in the interval of 18-25°C and statistically lower growth rates of 15, 28, and 30°C (Table VIII). Identical conclusions to these were drawn from regression analyses of mean weights of individuals in populations of 2, 4, and 8 earthworms in relation to time, where each population size was examined separately (data not shown). An analysis of variance showed that final mean weight achieved at 80 days was nonsignificant at 18, 20, and 25°C.

TABLE V
Total Live Weight (mg) of Singly Reared, or Populations of 2, 4, or 8 *E. foetida* after 30 Days in Relation to Temperature[a]

Temperatures (°C)	Number of worms per dish			
	1	2	4	8
	64	128	294	754
18	*340*	464	1077	*2064*
20	*378*	582	1256	*2870*
25	*446*	*1127*	*2160*	*2432*
28	*415*	*1300*	*1900*	*2522*
30	*368*	880	1625	1625

[a]Values are means based on 5 replicates. Italic values are significantly larger than other values, but statistically similar to each other, within columns.

TABLE VI
Mean Live Weight (mg) of *E. foetida* after 50 Days[a]

Temperature (°C)	Number of worms per dish			
	1	2	4	8
15	275	329	370	320
18	709	748	561	*554*
20	*1092*	838	*830*	(*676*)
25	796	(*1090*)	*826*	(*393*)
28	700	850	*750*	(*410*)
30	(522)	925	*725*	(*414*)

[a]Values of mean weight of individuals in populations calculated only for dishes in which total original number of worms per dish had survived from time of hatching. Significance (0.05) in italic and parenthesized for greater values within columns, and parenthesized for significantly smaller values within rows. Each number is based on at least 4 replicates, except at 30°C, where values are based on 4, 3, 4, and 2 replicates for 1, 2, 4, and 8 worms, respectively.

TABLE VII
Mean Live Weight (mg) of *E. foetida* after 80 Days[a]

Temperature (°C)	Number of worms per dish			
	1	2	4	8
15	*706**(5)	1018(4)	1018(3)	1088 (5)
18	1500 (5)	1252(4)	1252(3)	1188*(2)
20	1456 (4)	1330(4)	1330(5)	1310 (4)
25	1300 (5)	1350(3)	1350(5)	1325 (4)
28	1180 (5)	948(2)	1129(3)	1105 (2)
30	*720* (3)	*630*(1)	*630*(1)	518[b]

[a]Values based on number of replicates (parentheses) in which total original number of worms per dish survived from time of hatching. Significantly (0.05) low values are in italic and/or marked with an asterisk for comparisons within columns and rows, respectively.

[b]Based on 1 dish with 6 survivors.

TABLE VIII
Mean Live Weight (mg) of Surviving *E. foetida* in Relation to Time and Temperature[a]

Temperature (°C)	Time (days) 30	50	65	80	Regression equation[b]	r
15	74±7	323±19	619±46	937±115	Y = 11.7X − 136	0.94
18	275±24	643±50	*1072*±112	*1313*±94	Y = 17.0X − 106	0.98
20	335±20	*859*±86	*1186*±109	*1365*±45	Y = 18.1X − 67	0.99
25	*463*±59	*776*±194	*1050*±90	*1325*±14	Y = 16.4X − 19	0.99
28	*564*±70	*723*±121	786±80	795±95	Y = 9.9X + 127	0.93
30	354±52	646±113	622±65	622±43	Y = 8.2X + 78	0.93

[a]Values are $\bar{x} \pm$ SE of weight of 5 replicates of singly reared worms and mean weight 2, 4, and 8 worms per dish, 5 replicates of each dish.

[b]$P < 0.05$ for slopes at 15, 28, and 30°C relative to 18, 20, and 25°C.

Optimum Temperature for Total Production

Production of total biomass (cocoons and adults which survived 80 days) occurred optimally between 18 and 25°C, maximally so at 20°C, and with a precipitous dropoff above 25°C (Fig. 1).

DISCUSSION

Sexual maturity was reached earlier at 25°C than at 20°C, (Table II), but a similar number of cocoons was produced at both temperatures (Table IV). Development within cocoons occurred faster at 25°C than at 20°C—which confirms the work of Tsukamoto and Watanabe[8]—and although this might accelerate population growth at the higher temperature, significantly fewer hatchlings emerge at 25°C than at 20°C.[8,12] Regression analyses and analysis of variance of growth rates and final mean weight of individuals in populations of two, four, or eight earthworms revealed nonsignificant differences at 18, 20, and 25°C; this agrees with results for rate and production of total biomass (Table VIII). Difference in respiratory rate at 20 and 25°C was nonsignificant,[10,11] and slope between mean rates of weight loss during starvation was significantly smaller between 20 and 25°C than at 15-20°C and 25-30°C (unpublished experiment): the metabolic zone of thermal neutrality is thus the same for catabolism and anabolism.

All of the above, together with the fact that the source of food (microbial

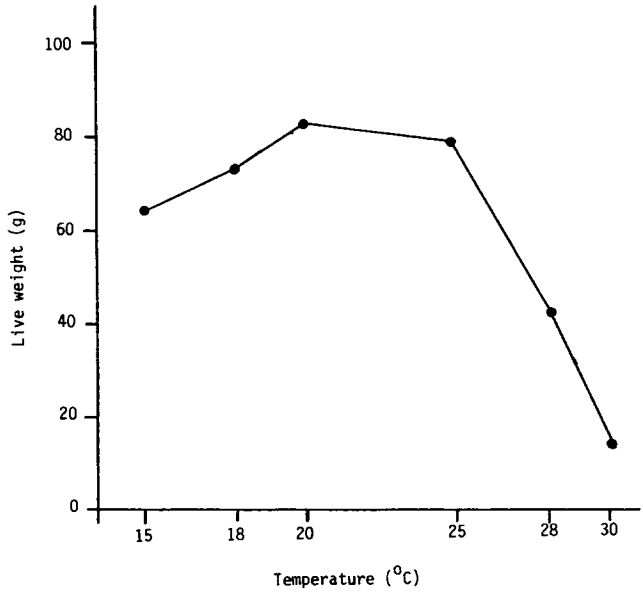

Fig. 1. Total biomass of *E. foetida*, including cocoons and adults, after 80 days, in relation to temperature. Values are based only on survivors. Ten milligrams was assigned as weight/cocoon.

biomass and cellulose) for *E. foetida* oxidizes less rapidly at 20°C than at 25°C leads to the tentative conclusion that 20°C is optimal for maximum production of *E. foetida* over a long period. In view of energy requirements for maintaining relatively constant ambient temperatures in warm versus cold months, operation at 25°C may be more favorable in the former, and at 20°C in the latter, months.[15]

EPILOGUE

Nearly all commercial worm growers in the United States market *E. foetida* as their product, though some individuals may also produce the exotic African nightcrawler *Eudrilus eugeniae* or pheretemoids of the genus *Amynthas*. The only marketplace for earthworms presently are bait shops (for fishing) and seed stores (for alleged improvement of garden soils).

Much of the basic science needed for commercial production of earthworms has been referred to recently.[7] Based on that report, it appears earthworms *could* be used in management of manures, where failure to do so in areas that are frostbound during much of the winter could bring the farmer into conflict with environmental law;[16] use of *E. foetida* might in fact provide a potentially economic solution to this problem.

Whether *E. foetida* can be used economically for protein production, however, depends primarily on a) whether low yields can be shown to be profitable, assuming a source of food is used once, and b) whether a source of food can be used repeatedly, thus obtaining multiple low yields which sum up to a profitable amount. Independently of whether the source of food contains 0.5 or 5% N, and a corresponding tenfold range of other essential mineral nutrients, yield is about 5 g dry *E. foetida* per initial 100 g dry food source.[17] (This ratio ignores whatever quantities of organic material disappear due to microbial metabolism.) This low yield of 5% is probably linked to castings toxicity,[18] wherein *E. foetida* will only consume a specific fraction of the food source in which it itself resides and into which it deposits its wastes.

The low yield perhaps militates against use of *E. foetida,* on economic grounds, for protein production. On the other hand, should favorable microbiological conditions exist for rapid detoxication of the spent medium (castings) following harvest of earthworms, and assuming a sufficient residue of microbial biomass and cellulose in the spent medium, a potential for economic production might exist. To realize this potential, hatchlings from the cocoons in the spent medium could be allowed to grow 8–10 weeks to obtain another 5%yield. Conceivably, this cycle of periodic harvest and growth upon spent medium could be repeated until microbial biomass, cellulose, or some other essential nutrient of *E. foetida*[19] becomes limiting.

All of this is in need of further research, but the possibility of rapidly converting straw into artificial manure,[20] the extensive research of Han and his colleagues on making straw more digestible to cattle through use of microorganisms,[21] and the possibility of enriching straw with earthworm protein prior to use as feedstock might justify additional work.

In a previous article, on production of earthworms as a potentially economical source of protein,[7] a net yield of about 2 g of live *E. foetida* were obtained on a 24 cm^2 area about 5 cm deep during 7 weeks between ages 20 and 27 weeks. This net yield was in the form of cocoons and hatchlings. In the present work, from Table VII, based on a mean of 1300 mg live weight per adult in populations of eight worms at 20 or 25°C, 10.4 g of biomass were produced in 80 days in an area 78 cm^2, 1.5-2 cm deep. This converts to approximately the same level of productivity as in the preceding study, but is based on a net yield in the form of adult biomass. *E. foetida* is a potentially rich source of protein, but whether this potential can be realized hinges on obtaining further insight into its specific nutrient requirements, on achieving conditions for optimizing the yield from a starting source of food, and on whether engineering can be devised for appropriate sequencing of operations in which cocoons or hatchlings would be introduced and adults harvested.

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